QUESTION ONE: ENERGY CHANGES

A drone releases a small metal payload of mass **0.320 kg** from a height of **42.0 m**. The payload is equipped with a small parachute that opens **0.8 seconds** after release. Before the parachute opens, air resistance is negligible. After it opens, air resistance becomes significant and varies with speed.

(a)

Analyse the energy transformations from release until the payload reaches the ground.

Your answer must include:

- How gravitational potential energy, kinetic energy, and thermal energy interact
- How the *timing* of the parachute opening affects the energy distribution
- Why the total mechanical energy is not conserved

(b)

Calculate the speed of the payload **at the moment the parachute opens**, assuming no air resistance before opening.

(c)

After the parachute opens, the payload reaches a constant terminal velocity of **6.8 m/s**. Analyse why terminal velocity occurs, using energy concepts and force balance.

Your answer must explain:

- Why kinetic energy stops increasing
- Where the gravitational energy continues to go
- Why terminal velocity is lower with a parachute than without one

(d)

The payload hits the ground at **6.8 m/s**.
Calculate the total mechanical energy lost due to air resistance from the moment the parachute opens until impact.

Then analyse **why** this value is significantly larger than the energy lost before the parachute opened.

(e)

The drone operator now attaches a *larger* parachute.

Predict and analyse how this change affects:

- Terminal velocity
- Time to reach the ground

Total mechanical energy lost

- Your answer must include **energy reasoning**, not just force reasoning.

QUESTION TWO: THERMAL ENERGY

A sealed metal cylinder contains **0.950 kg** of water at **18°C**. A heating element supplies **120 W** of power.
The cylinder itself absorbs **22%** of the supplied energy.
The specific heat capacity of water is **4200 J kg^{-1} °C^{-1}**.

The cylinder is perfectly sealed, so no steam can escape.

(a)

Analyse the meaning of *thermal energy* in this context, and explain how it relates to both molecular motion and macroscopic temperature change.

(b)

Calculate the time required for the water to reach **100°C**.

(c)

Once the water reaches **100°C**, the temperature continues to rise to **118°C** because the cylinder is sealed.

Analyse why water can exceed 100°C in this situation, referring to:

- Pressure
- Boiling point

Energy distribution

- Your answer must include a clear explanation of why boiling does not occur normally.

(d)

At **118°C**, the water begins to boil under pressure.
Explain why the temperature remains constant during boiling even though energy is still being supplied.

Your answer must reference **latent heat**, **phase change**, and **internal energy**.

(e)

The cylinder is suddenly opened, releasing steam explosively.

Analyse the energy changes that occur during this rapid depressurisation, including:

- Why the water temperature instantly drops
- Why steam rapidly expands

How energy is redistributed between thermal, kinetic, and internal energy

- Your answer must show deep conceptual understanding.

QUESTION THREE: ELECTRICITY

A circuit contains a 24 V power supply, a heating coil, a fixed resistor, and a variable resistor.
The heating coil has resistance **10.0 Ω**.
The fixed resistor has resistance **6.0 Ω**.
The variable resistor can be adjusted between **0–30 Ω**.

All components are connected in series.

(a)

Describe (in words) where an ammeter and voltmeter should be placed to measure:

- Total circuit current

Voltage across the heating coil

- Then analyse **why** incorrect placement leads to meaningless or distorted readings.

(b)

The variable resistor is set to **18 Ω**.

Calculate:

- The total circuit resistance
- The circuit current

The electrical power delivered *specifically* to the heating coil

- Your answer must include reasoning about how power is distributed in a series circuit.

(c)

The heating coil is submerged in oil.

Analyse why oil is used instead of water or air, referring to:

- Thermal conductivity
- Heat capacity
- Electrical safety
- Energy transfer rate

(d)

The variable resistor is increased from **18 Ω** to **30 Ω**.

Analyse how this affects:

- Circuit current
- Voltage across the heating coil

Heating rate of the oil

- Your answer must reference **Ohm's law**, **energy transfer**, and **power relationships**.

(e)

The heating coil is replaced with a coil of **higher resistance**.

Analyse how this change affects:

- Current
- Voltage distribution

- Power delivered to the coil

Heating effectiveness

- Your answer must include **quantitative reasoning** and **energy concepts**, not just statements.